

ES — Elements from the Sea

OCR B (Salters) H433 storyline **ES**. Chemistry from sea water: halogen chemistry, redox and electrolysis, the introduction of equilibrium, and atom economy.

1. Oxidation States and Redox Reactions

Oxidation States (Oxidation Numbers)

Oxidation state: A number assigned to an atom in a compound that represents the number of electrons it has lost, gained, or shared unequally compared to the uncombined element. It is a bookkeeping device for tracking electron transfer.

Rules for assigning oxidation states:

- Uncombined elements have an oxidation state of 0.
- In simple ions, the oxidation state equals the charge (e.g. $\text{Na}^+ = +1$, $\text{Cl}^- = -1$).
- Oxygen is -2 (except in peroxides where it is -1 , and in OF_2 where it is $+2$).
- Hydrogen is $+1$ (except in metal hydrides where it is -1).
- Fluorine is always -1 .
- The sum of oxidation states in a neutral compound is 0; in a polyatomic ion it equals the ionic charge.

Redox Definitions

Oxidation: Loss of electrons / increase in oxidation state.

Reduction: Gain of electrons / decrease in oxidation state.

Reducing agent: A substance that donates electrons (is itself oxidised).

Oxidising agent: A substance that accepts electrons (is itself reduced).

Remember: OIL RIG — Oxidation Is Loss, Reduction Is Gain (of electrons).

Half-Equations

A redox reaction can be split into two half-equations: one showing oxidation and one showing reduction. To write and balance half-equations:

1. Write the species being oxidised or reduced on each side.
2. Balance atoms other than O and H.
3. Balance oxygen by adding H_2O .
4. Balance hydrogen by adding H^+ .
5. Balance charge by adding electrons (e^-).

To combine half-equations, multiply each so the electrons cancel, then add the two equations together and simplify.

Exam Tip: When balancing redox equations, always check that atoms and charges balance on both sides. A common mistake is forgetting to balance hydrogen atoms after adding water molecules. Also check that the electrons cancel completely when combining half-equations.

2. Dynamic Equilibrium

Dynamic equilibrium: A state reached in a closed system when the rate of the forward reaction equals the rate of the reverse reaction, so the concentrations of reactants and products remain constant but are not necessarily equal. Both forward and reverse reactions continue to occur at the molecular level.

Conditions for dynamic equilibrium:

- The system must be closed (no substances can enter or leave).
- The forward and reverse rates must be equal.

- The concentrations of all species remain constant over time.
- The reaction must be reversible.

Le Chatelier's Principle

Le Chatelier's principle: If a system at equilibrium is subjected to a change in conditions (concentration, pressure, or temperature), the position of equilibrium will shift in the direction that tends to oppose the change, so as to partially restore the original conditions.

Change	Effect on equilibrium position	Effect on K
Increase concentration of a reactant	Shifts to the right (towards products)	No change
Decrease concentration of a product	Shifts to the right	No change
Increase pressure	Shifts towards the side with fewer moles of gas	No change
Increase temperature	Shifts in the endothermic direction	Changes (increases if forward reaction is endothermic)
Add a catalyst	No change (speeds up both forward and reverse equally)	No change

Exam Tip: A catalyst does not change the position of equilibrium or the value of K. It only increases the rate at which equilibrium is reached. Temperature is the only factor that changes the value of the equilibrium constant.

Le Chatelier's Principle — Equilibrium Shifts



← Reactants

Equilibrium Position

Products →

Increase $[\text{N}_2]$

Shifts right

Increase pressure

Shifts right
(fewer gas moles)

Increase temperature

Shifts left (endo)

Add catalyst

NO SHIFT

Faster only

Concentration & pressure changes: K unchanged, position shifts to oppose change
 Temperature changes: K changes, shifts in the endothermic direction

3. Halogen Chemistry

Sea water is a halide solution, and the halogens are the industrial products won from it. Group 17 elements have the outer configuration ns^2np^5 , are one electron short of a noble gas configuration, and exist as diatomic molecules X_2 .

Physical Properties

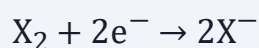
Halogen	Appearance and state at room temperature	Boiling point / °C	Colour in water	Colour in cyclohexane
F_2	Pale yellow gas	-188	Reacts with water	—
Cl_2	Pale green gas	-34	Very pale green	Pale green / almost colourless
Br_2	Red-brown liquid	59	Orange / yellow	Orange
I_2	Grey-black solid, sublimes to a purple vapour	184	Barely soluble; pale brown	Violet / purple

Boiling points rise and volatility falls down the group. The molecules are non-polar, so the only intermolecular bonds are instantaneous dipole–induced dipole bonds. Going down the group each molecule has more electrons, so the induced dipoles are larger, these bonds are stronger, and more energy is needed to separate the molecules.

The halogens are only slightly soluble in water (they are non-polar) but dissolve readily in non-polar organic solvents such as cyclohexane. Shaking an aqueous halogen with cyclohexane and looking at the colour of the upper organic layer is the standard way of telling bromine from iodine.

The Halogens as Oxidising Agents

A halogen reacts by gaining one electron per atom:



Oxidising power **decreases** down the group. Down the group the atomic radius increases and there is more shielding, so the nucleus attracts an incoming electron less strongly and the halogen gains electrons less readily. Chlorine is therefore a stronger oxidising agent than bromine, which is stronger than iodine.

Displacement Reactions of the Halogens

A more reactive halogen displaces a less reactive halogen from a solution of its halide. These are the reactions to know, with observations:

Mixture	Observation	Ionic equation
$\text{Cl}_2(\text{aq}) + \text{KBr}(\text{aq})$	Colourless solution turns orange	$\text{Cl}_2 + 2\text{Br}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$
$\text{Cl}_2(\text{aq}) + \text{KI}(\text{aq})$	Colourless solution turns brown; a dark solid may appear	$\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2$
$\text{Br}_2(\text{aq}) + \text{KI}(\text{aq})$	Orange solution turns brown	$\text{Br}_2 + 2\text{I}^- \rightarrow 2\text{Br}^- + \text{I}_2$
$\text{Br}_2(\text{aq}) + \text{KCl}(\text{aq})$	No change	No reaction

$\text{I}_2(\text{aq}) + \text{KCl}(\text{aq})$ or $\text{KBr}(\text{aq})$	No change	No reaction
--	-----------	-------------

Each is a redox reaction. Taking chlorine and bromide as the example, the half-equations are:



Chlorine is the oxidising agent (it is reduced, -1 from 0); the bromide ion is the reducing agent (it is oxidised, 0 from -1).

Extracting the Halogens from the Sea

Sea water contains about 65 parts per million of bromide ion. Bromine is recovered by exploiting exactly the displacement chemistry above:

1. The sea water is acidified and chlorine is blown through it, oxidising bromide to bromine: $\text{Cl}_2 + 2\text{Br}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$
2. Air is blown through to carry the volatile bromine out of the very dilute solution.
3. The bromine vapour is absorbed in sulfur dioxide and water, which reduces it back to bromide and concentrates it: $\text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{HBr} + \text{H}_2\text{SO}_4$
4. Chlorine is used again on the concentrated solution to release pure bromine: $\text{Cl}_2 + 2\text{HBr} \rightarrow 2\text{HCl} + \text{Br}_2$

Chlorine itself is made by the electrolysis of concentrated sodium chloride solution (brine), covered in the next section.

Exam Tip: The bromine extraction looks circular — bromine is made, turned back into bromide, then made again. The point of the middle two steps is *concentration*: bromine in sea water is far too dilute to collect directly, so it is transferred into a small volume of solution before the final oxidation. Say that explicitly if you are asked to explain the process.

Testing for Halide Ions with Silver Nitrate

Add dilute nitric acid (to remove carbonate and hydroxide ions, which would also give precipitates) and then silver nitrate solution:

Halide	Precipitate	Ionic equation	Dilute NH ₃	Concentrated NH ₃
Cl ⁻	White	$\text{Ag}^+ + \text{Cl}^- \rightarrow \text{AgCl(s)}$	Dissolves	Dissolves
Br ⁻	Cream	$\text{Ag}^+ + \text{Br}^- \rightarrow \text{AgBr(s)}$	Stays	Dissolves
I ⁻	Yellow	$\text{Ag}^+ + \text{I}^- \rightarrow \text{AgI(s)}$	Stays	Stays

The three precipitate colours are hard to tell apart in poor light, which is why the ammonia step matters: solubility in ammonia decreases from AgCl to AgI, so the sequence of tests identifies the halide unambiguously. Silver halides darken in sunlight as Ag⁺ is reduced to silver metal — the chemistry behind photographic film.

The Hydrogen Halides

Adding *concentrated* sulfuric acid to a solid halide shows the reducing power of the halide ions increasing down the group.

Solid halide	What happens	Equations	Observations
NaCl	Acid–base only. Cl ⁻ is too weak a reducing agent to reduce sulfuric acid.	$\text{NaCl} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HCl}$	Steamy, misty fumes of HCl
NaBr	Acid–base, then redox: HBr reduces S from +6 to +4.	$\text{NaBr} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HBr}$ $2\text{HBr} + \text{H}_2\text{SO}_4 \rightarrow \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$	Steamy fumes, then orange-brown bromine vapour and choking SO ₂
NaI	Acid–base, then strong redox: HI reduces S all the way from +6 to -2.	$\text{NaI} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HI}$ $8\text{HI} + \text{H}_2\text{SO}_4 \rightarrow 4\text{I}_2 + \text{H}_2\text{S} + 4\text{H}_2\text{O}$	Purple iodine vapour, black solid, yellow sulfur, and H ₂ S smelling of bad eggs

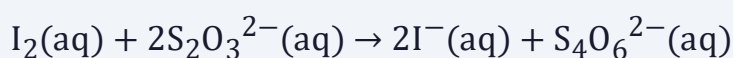
Because sulfuric acid oxidises HBr and HI, it cannot be used to prepare them. Instead the solid halide is treated with **phosphoric acid**, which is not an oxidising agent, so pure HBr or HI is obtained.

The hydrogen halides share two properties and differ in a third:

- **Acidity (similar):** all dissolve in water to give strongly acidic solutions, e.g. $\text{HCl(g)} + \text{H}_2\text{O(l)} \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{Cl}^-(\text{aq})$.
- **Reaction with ammonia (similar):** all give dense white smoke of the ammonium halide, e.g. $\text{NH}_3(\text{g}) + \text{HCl(g)} \rightarrow \text{NH}_4\text{Cl(s)}$.
- **Thermal stability (different):** decreases down the group, because the H–X bond gets longer and weaker. HCl is essentially unaffected by a hot wire; HBr partly decomposes; HI decomposes readily to give purple iodine vapour.

Iodine–Thiosulfate Titrations

Iodine is a convenient thing to titrate because the end point is so sharp. Sodium thiosulfate reduces iodine back to iodide:



Thiosulfate is run in from the burette until the brown iodine colour fades to pale straw. *Only then* is starch solution added, turning the mixture blue-black, and the titration is finished dropwise until the blue-black colour disappears. Starch is added late because it forms a stubborn complex with iodine at high concentration, which would make the end point drift.

Worked Example — Iodine–Thiosulfate Titration:

A 25.0 cm^3 portion of an iodine solution required 22.40 cm^3 of $0.100 \text{ mol dm}^{-3}$ sodium thiosulfate solution to reach the end point. Calculate the concentration of the iodine solution.

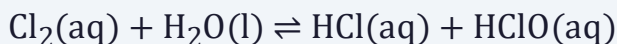
$$n(\text{S}_2\text{O}_3^{2-}) = 0.100 \times 22.40/1000 = 2.24 \times 10^{-3} \text{ mol}$$

The equation shows 2 mol thiosulfate per 1 mol iodine, so $n(\text{I}_2) = 2.24 \times 10^{-3} / 2 = 1.12 \times 10^{-3} \text{ mol}$

$$c(\text{I}_2) = 1.12 \times 10^{-3} / (25.0/1000) = \mathbf{0.0448 \text{ mol dm}^{-3}}$$

Chlorine: Weighing Risks Against Benefits

Chlorine dissolves in and reacts with water in a disproportionation reaction — chlorine is simultaneously oxidised to +1 and reduced to −1:



Chloric(I) acid, HClO, is the species that kills bacteria. With cold dilute sodium hydroxide the same disproportionation gives household bleach:



Risks	Benefits
Toxic gas that attacks the lungs; used as a chemical weapon in the First World War	Sterilising drinking water has saved enormous numbers of lives from cholera and typhoid
Transported as a liquid under pressure, so a tanker accident could release a dense cloud that hugs the ground	Bleaching paper and textiles
Can form chlorinated organic by-products in treated water	Feedstock for PVC, solvents and many medicines

The judgement the storyline asks for is not "chlorine is dangerous" but a comparison: the very small risk from a carefully regulated supply chain set against a very large public-health benefit, and against the risks of the alternatives.

Naming Inorganic Compounds

Where an element can have more than one oxidation state, the state is given as a Roman numeral in brackets immediately after the element it refers to.

Formula	Name	Reasoning
Cu ₂ S	Copper(I) sulfide	S is −2, shared over two Cu, so each Cu is +1

NaClO	Sodium chlorate(I)	Na +1, O -2, so Cl is +1
Pb(NO ₃) ₂	Lead(II) nitrate(V)	Two nitrate ions of charge -1, so Pb is +2; in NO ₃ ⁻ , N is +5
KMnO ₄	Potassium manganate(VII)	K +1, four O at -2, so Mn is +7
NaClO ₃	Sodium chlorate(V)	Na +1, three O at -2, so Cl is +5

4. Electrolysis

Electrolysis: The decomposition of a molten or dissolved ionic compound by passing a direct electric current through it. The compound being decomposed is the **electrolyte**; it must contain ions that are free to move.

The two electrodes do opposite jobs, and the names are worth fixing once:

Electrode	Charge	Attracts	Process	Memory hook
Cathode	Negative	Cations (positive ions)	Reduction — ions gain electrons	Cathode, Cations, reduCtion
Anode	Positive	Anions (negative ions)	Oxidation — ions lose electrons	Anode, Anions, oxidAtion

Electrolysis of Molten Salts

With no water present, the only ions available are those of the salt itself, so the products are simply the element from each ion.

Molten sodium chloride:



Molten aluminium oxide (dissolved in molten cryolite to lower the operating temperature):



Electrolysis of Aqueous Solutions

In solution the ions of the salt have to compete with water, and the rules are:

Electrode	Rule	Half-equation
Cathode	If the metal is in Group 1 or Group 2, or is aluminium, hydrogen is given off from water instead of the metal	$2\text{H}_2\text{O} + 2\text{e}^{-} \rightarrow \text{H}_2 + 2\text{OH}^{-}$
	Any other metal is plated onto the electrode	e.g. $\text{Cu}^{2+} + 2\text{e}^{-} \rightarrow \text{Cu}$
Anode	If a halide ion is present in reasonable concentration, the halogen is released	$2\text{Cl}^{-} \rightarrow \text{Cl}_2 + 2\text{e}^{-}$
	Otherwise (sulfate, nitrate) oxygen is released from water	$2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^{+} + 4\text{e}^{-}$

Concentrated sodium chloride solution (brine) is the industrially important case, and it is where the chlorine of this storyline comes from:



Sodium ions and hydroxide ions are left behind, so the chlor-alkali process yields three saleable products from sea salt and electricity: chlorine, hydrogen and sodium hydroxide. A modern membrane cell uses an ion-exchange membrane that lets Na^{+} through but keeps the chlorine and the hydroxide apart — if they mixed they would react to give bleach rather than the two separate products.

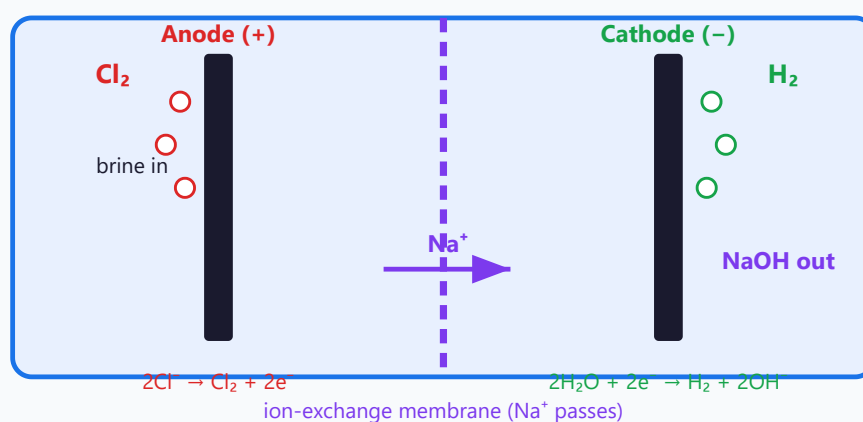
Copper(II) sulfate solution with inert (graphite or platinum) electrodes:



Copper is not in Group 1 or 2, so it plates out; sulfate is not a halide, so oxygen comes off instead. The solution loses its blue colour and becomes acidic as H^+ builds up.

Exam Tip: Electrons never appear in an overall equation, only in half-equations. When you combine them, scale each half-equation so the electrons cancel exactly — for brine, the cathode and anode half-equations both involve 2e^- , so they add directly to give $2\text{Cl}^- + 2\text{H}_2\text{O} \rightarrow \text{Cl}_2 + \text{H}_2 + 2\text{OH}^-$. Always check that charge as well as atoms balances in a half-equation.

Electrolysis of Brine — the Membrane Cell



Three products from salt and electricity: Cl_2 , H_2 and NaOH